Math 1232 Summer 2025
Practice Midterm

- You will have 90 minutes for this test.
- You are not allowed to consult books or notes during the test, but you may use a one-page, one-sided, handwritten cheat sheet you have made for yourself ahead of time.
- You may not use a calculator. You may leave answers unsimplified, except you should compute trigonometric functions as far as possible.
- The exam has 6 problems, one on each mastery topic we've covered. The exam has 6 pages total.
- Each part of each topic is worth ten points. The whole test is scored out of 100 points.
- Read the questions carefully and make sure to answer the actual question asked. Make sure to justify your answers—math is largely about clear communication and argument, so an unjustified answer is much like no answer at all. When in doubt, show more work and write complete sentences.
- If you need more paper to show work, I have extra at the front of the room.
- Good luck!

	a	b	c
M1:			
M2:			
S1:		S2:	
S3:		S4:	

Total ___ /100

Name: Solutions

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Problem 1 (M1).

(a) Compute $\frac{d}{dx}\sin(x)^{(x^2+1)}$.

Solution.

$$\begin{aligned} y &= \sin(x)^{x^2+1} \\ \ln(y) &= (x^2+1)\ln(\sin(x)) \\ \frac{y'}{y} &= 2x\ln(\sin(x)) + (x^2+1)\frac{\cos(x)}{\sin(x)} \\ y' &= y\left(2x\ln(\sin(x)) + (x^2+1)\frac{\cos(x)}{\sin(x)}\right) \\ &= \sin(x)^{x^2+1}\left(2x\ln(\sin(x)) + (x^2+1)\frac{\cos(x)}{\sin(x)}\right) \end{aligned}$$

(b) Compute $\displaystyle\int x^5 + 5^x\,dx$.

Solution.

$$\int x^5 + 5^x\,dx = \frac{x^6}{6} + \frac{5^x}{\ln(5)} + C.$$

(c) Compute $\displaystyle\int \frac{1}{x\sqrt{4-\ln(x)^2}}\,dx$.

Solution. We can set $u = \ln(x)/2$ so $du = \frac{1}{2x}\,dx$. Then

$$\begin{aligned} \int \frac{1}{x\sqrt{4-\ln(x)^2}}\,dx &= \int \frac{2}{\sqrt{4-4u^2}}\,du = \int \frac{1}{\sqrt{1-u^2}}\,du \\ &= \arcsin(u) + C = \arcsin(\ln(x)/2) + C. \end{aligned}$$

Name: Solutions

Problem 2 (M2). Compute the following integrals using any of the methods we have learned in class. Show enough work to justify your answers.

(a) $\displaystyle\int \sin(2x)e^{5x}\,dx$.

Solution.

$$\begin{aligned}
\int \sin(2x)e^{5x}\,dx &= \sin(2x)\frac{e^{5x}}{5} - \int 2\cos(2x)\frac{e^{5x}}{5}\,dx \\
&= \frac{1}{5}\sin(2x)e^{5x} - \frac{2}{5}\left(\cos(2x)\frac{e^{5x}}{5} - \int -2\sin(2x)\frac{e^{5x}}{5}\,dx\right) \\
&= \frac{1}{5}\sin(2x)e^{5x} - \frac{2}{25}\cos(2x)e^{5x} + \frac{4}{25}\int \sin(2x)e^{5x}\,dx \\
\frac{29}{25}\int \sin(2x)e^{5x}\,dx &= \frac{1}{5}\sin(2x)e^{5x} - \frac{2}{25}\cos(2x)e^{5x} + C \\
\int \sin(2x)e^{5x}\,dx &= \frac{5}{29}\sin(2x)e^{5x} - \frac{2}{29}\cos(2x)e^{5x} + C.
\end{aligned}$$

(b) $\displaystyle\int \frac{x^4+1}{x^3+4x}\,dx$

Solution. Polynomial long division gives that

$$\frac{x^4+1}{x^3+4x} = x + \frac{1-4x^2}{x^3+4x} = x + \frac{1-4x^2}{x(x^2+4)}$$

Decomposing, we look for A, B, C with

$$\frac{1-4x^2}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4} \iff 1-4x^2 = A(x^2+4) + (Bx+C)x$$

Substituting $x=0$ gives $4A=1$ so $A=1/4$; $x=1$ gives $-3 = 5A+B+C = 5/4+B+C$; $x=-1$ gives $-3 = 5/4+B-C$. We have the system of equations $B+C=-17/4$, $B-C=-17/4$, so $B=-17/4$ and $C=0$. Thus

$$\begin{aligned}
\int \frac{x^4+1}{x^3+4x}\,dx &= \int x + \frac{1}{4x} - \frac{17x}{4(x^2+4)}\,dx \\
&= \frac{x^2}{2} + \frac{1}{4}\ln|4x| - \frac{17}{8}\ln|x^2+4| + C.
\end{aligned}$$

Name: Solutions

(c) $\displaystyle\int_0^{1/\sqrt{3}} \frac{x^3}{\sqrt{1+x^2}}\,dx$

Solution. We'll set $x = \tan(\theta)$, and then $dx = \sec^2(\theta)\,d\theta$. As x goes from 0 to $1/\sqrt{3}$ we have $\tan\theta$ going from 0 to $1/\sqrt{3}$, and thus θ going from 0 to $\pi/6$. Then we have

$$\begin{aligned}
\int_0^{1/\sqrt{3}} \frac{x^3}{\sqrt{1+x^2}}\,dx &= \int_0^{\pi/6} \frac{\tan^3(\theta)}{\sqrt{1+\tan^2(\theta)}}\sec^2(\theta)\,d\theta = \int_0^{\pi/6} \tan^3(\theta)\sec(\theta)\,d\theta \\
&= \int_0^{\pi/6} \tan(\theta)\sec(\theta)(\sec^2(\theta)-1)\,d\theta = \frac{1}{3}\sec^3(\theta) - \sec(\theta)\Big|_0^{\pi/6} \\
&= \frac{8}{9\sqrt{3}} - \frac{2}{\sqrt{3}} - \left(\frac{1}{3} - 1\right) = \frac{2}{3} - \frac{10}{9\sqrt{3}}.
\end{aligned}$$

Name: Solutions

Problem 3 (S1). Is $f(x) = e^{x^2}$ invertible or not? Justify your answer.

Solution. We have $f(-1) = f(1) = e$ so this function is not one-to-one, and thus not invertible.

Problem 4 (S2). Compute $\lim_{x \to 0} (2x+1)^{\cot(x)}$.

Solution. Taking logs of both sides gives

$$\begin{aligned} y &= (2x+1)^{\cot(x)} \\ \ln|y| &= \cot(x)\ln|2x+1| = \frac{\cos(x)\ln|2x+1|}{\sin(x)}. \end{aligned}$$

The top and bottom both approach 0, so we can use L'Hôpital's Rule:

$$\begin{aligned} \lim_{x \to 0} \ln|y| &= \lim_{x \to 0} \cos(x)\frac{\ln|2x+1|}{\sin(x)} = 1 \cdot \lim_{x \to 0} \frac{\ln|2x+1|}{\sin(x)} \\ &\overset{\text{L'H}}{=} \lim_{x \to 0} \frac{2/(2x+1)}{\cos(x)} = 2 \\ \lim_{x \to 0} y &= e^2. \end{aligned}$$

Name: Solutions

Problem 5 (S3). Compute $\displaystyle\int_1^2 \frac{dx}{x\ln(x)}$.

Solution.

$$\begin{aligned}\int_1^2 \frac{dx}{x\ln(x)} &= \lim_{t\to 1^+}\int_t^2 \frac{dx}{x\ln(x)}\\ &= \lim_{t\to 1^+} \ln(|\ln(x)|)\Big|_t^2\\ &= \lim_{t\to 1^+} \ln(|\ln(2)|) - \ln|\ln(t)|\end{aligned}$$

But $\lim_{t\to 1^+}\ln(t) = 0$, so $\lim_{t\to 1^+}\ln|\ln(t)| = -\infty$. So this integral diverges.

Problem 5 (S4). Compute the arc length of the curve $(y-2)^3 = x^2$ between $y = 2$ and $y = 6$ for $x \geq 0$.

Solution. We have $x = (y-2)^{3/2}$, so $\frac{dx}{dy} = \frac{3}{2}(y-2)^{1/2}$ and

$$\begin{aligned}L &= \int_2^6 \sqrt{1+\frac{9}{4}(y-2)}\,dy\\ &= \frac{8}{27}\left(1+\frac{9}{4}(y-2)\right)^{3/2}\Bigg|_2^6\\ &= \frac{8}{27}\left(10^{3/2}-1\right).\end{aligned}$$